

Final Project: Integrating Your ML Pipeline

Part B: Technical Report PDF

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B.1: Introduction, Problem Description, and Motivation

Our team's project focuses on the challenge of multiclass text classification using the HuffPost news dataset (Misra, 2018), aiming to categorize news articles into distinct categories based on their headlines and short descriptions. In a plausible business context, such as a high-volume digital news aggregator or an automated content management system, this model serves a critical function by enabling automated topic routing. Instead of relying on manual tagging by editors, which is labor-intensive, slow, and difficult to scale, our solution allows an organization to instantly categorize thousands of incoming wire stories. This capability ensures that articles are accurately distributed to specific user interest feeds, such as "Wellness," "Politics," or "Technology," thereby enhancing user engagement and operational efficiency.

Project Goals: Business and Data Science

From a business perspective, the primary goal is to support the outcome of automated content curation, reducing operational bottlenecks associated with manual classification and ensuring timely content distribution. In data science terms, the model attempts to predict the correct category for a given text input from a set of 28 consolidated classes, optimized from an original set of 41 to create distinct decision boundaries. This prediction task operates under significant constraints, specifically the need to maximize generalization for classification performance, measured by Test Accuracy and Macro-F1 scores, while navigating severe class imbalance and overlap within the labels. The technical objective involves leveraging transfer learning with the RoBERTa architecture (Encoder) and optimizing the hyperparameters, such as Optimizer, learning rates, batch sizes, and layers to unfreeze and train, to achieve a robust classifier capable of handling these real-world data imperfections.

Dataset Description

The dataset utilized is the HuffPost News Category Dataset (Misra, 2018), sourced directly from Hugging Face, which consists of text records containing headlines and short descriptions. Originally comprising 41 distinct categories, the dataset exhibits severe class imbalance; dominant classes like "POLITICS" contain over 32,000 samples, whereas minority topics like "EDUCATION" have fewer than 1,000 records. Key input modalities are strictly textual, processed through a pipeline that concatenates headlines and descriptions while preserving contextual markers like punctuation and stop words for context and attention-based analysis. The project faced significant challenges regarding label noise and class overlap, evidenced by high TF-IDF similarity scores between categories such as "PARENTING" and "PARENTS" (0.849 similarity) or "FOOD & DRINK" and "TASTE" (0.852 similarity). To address this, we employed classification confusion patterns and cosine similarity n-gram (1,2) analysis to merge ambiguous classes, resulting in a final distribution of 28 consolidated categories for the training process based on TF-IDF similarity scores and confusion patterns.

Test accuracy: 0.7328
Test macro-F1: 0.6605

Per-Category Performance (Sorted by F1 Score)

WEIRD NEWS	0.407	406.0
IMPACT	0.432	345.0
WOMEN	0.462	339.0
LATINO VOICES	0.505	113.0
MEDIA	0.513	281.0
EDUCATION	0.515	214.0
COMEDY	0.532	511.0
BLACK VOICES	0.571	453.0
BUSINESS	0.590	763.0
TECH	0.592	421.0
ARTS & CULTURE	0.596	387.0
RELIGION	0.609	254.0
GREEN	0.610	393.0
STYLE	0.616	224.0
CRIME	0.639	340.0
THE WORLDPOST	0.732	841.0
ENTERTAINMENT	0.747	1605.0
QUEER VOICES	0.749	631.0
PARENTING	0.760	1255.0
SPORTS	0.769	488.0
HOME & LIVING	0.772	415.0
DIVORCE	0.788	342.0
WELLNESS	0.807	2586.0
POLITICS	0.809	3251.0
WEDDINGS	0.825	365.0
TRAVEL	0.839	988.0
FOOD & DRINK	0.847	832.0
STYLE & BEAUTY	0.863	950.0

Top misclassifications (True label -> Predicted label)

MEDIA	-> POLITICS	25.3%
COMEDY	-> ENTERTAINMENT	23.7%
BLACK VOICES	-> ENTERTAINMENT	19.6%
STYLE	-> STYLE & BEAUTY	15.6%
WOMEN	-> WELLNESS	15.3%
IMPACT	-> WELLNESS	13.0%
STYLE	-> ENTERTAINMENT	12.9%
MEDIA	-> ENTERTAINMENT	12.8%
ARTS & CULTURE	-> ENTERTAINMENT	12.4%
EDUCATION	-> PARENTING	12.1%

Stats collected: 1 runs after final training call.

Figure 1 and 2: Illustrate the count and labels

B.2: Methodology

Preprocessing Pipeline:

Our team prepared a data cleaning and normalization pipeline designed to transform raw JSON text into high-quality inputs for transformer-based learning. We began by concatenating the headline and short_description fields to maximize the semantic context available for classification. For text preprocessing pipeline, we utilized SpaCy (en_core_web_sm) to tokenize the text while explicitly choosing to preserve stop words and punctuation; this was a strategic pivot from our Milestone 2 approach, as we determined that transformer architectures require full sentence structures to effectively learn contextual embeddings. We also employed BeautifulSoup to strip HTML tags and regular expressions to remove URLs, as models are unable to understand these patterns. To ensure data quality, we filtered out records with fewer than 3 tokens and performed deduplication to remove exact text matches and ambiguous entries where identical text mapped to conflicting labels.

An important component of our methodology was Label Engineering to address severe class imbalance and semantic noise. We utilized TF-IDF vectorization (n-gram range 1-2) and cosine similarity analysis to quantify category overlap. This analysis revealed high similarity between categories such as "TASTE" and "FOOD & DRINK" (0.852 similarity) and "PARENTS" vs. "PARENTING" (0.849 similarity). Based on these metrics, we merged overlapping classes, reducing the target space from 41 to 28 consolidated categories. For the final model input, we used the KerasNLP RobertaTextClassifierPreprocessor with a maximum sequence length of 75 tokens, a value calculated to cover the 99th percentile of the dataset distribution.

Model architecture and Justification

We selected RoBERTa Base (Robustly Optimized BERT Pretraining Approach) as our final classifier. This decision was informed by our Milestone 2 and Final Milestone experiments, where simple baselines and custom architectures (BiLSTM) failed to capture the necessary semantic depth, achieving significantly lower accuracy. In our final comparative evaluation, RoBERTa consistently outperformed other transfer learning candidates, such as BERT and DistilBERT, offering superior training stability and higher validation accuracy. Based on the classification accuracy of the optimized models, we implemented a Partial Fine-Tuning strategy, unfreezing only the top 2 transformer layers (out of 12) while keeping the rest of the backbone frozen. This approach allowed the model to adapt to the specific nuances of the HuffPost syntax while retaining its pre-trained linguistic knowledge, offering an optimal balance between performance and computational efficiency. The architecture included a dropout rate of 0.5 on the classification head along with a dropout rate of 0.3 in the unfrozen later to further mitigate overfitting.

Training strategy

The model was trained using Sparse Categorical Crossentropy loss, appropriate for our integer-encoded multiclass targets. We utilized the Adam optimizer, which our Optuna hyperparameter search identified as the best option (compared to AdamW, SGD, and RMSprop). The training process used a batch size of 16 and a precise learning rate of $1.22e-5$, both optimized via the Tree-structured Parzen Estimator (TPE) sampler in Optuna. To handle the remaining class imbalance (where "POLITICS" significantly outnumbered classes like "EDUCATION"), we computed and applied balanced class weights during training. We employed robust callbacks to ensure convergence, including Early Stopping (patience of 5 epochs) to halt training when validation loss diverged, and ReduceLROnPlateau (factor 0.5,

patience 2), which dynamically lowered the learning rate to settle into a deeper loss minimum. The final model converged rapidly, with optimal weights restored from Epoch 3.

Performance validation

To evaluate performance and prevent data leakage, we split the dataset into 80% Training, 10% Validation, and 10% Test sets. Crucially, we used stratified splitting to ensure that the distribution of all 28 categories remained consistent across all three subsets, preventing representation bias for minority classes. The validation set was used exclusively for hyperparameter tuning, while the test set was completely held out until the final inference stage to provide an unbiased metric of real-world performance. Data leakage was further prevented by the initial deduplication step, ensuring no identical stories appeared in both training and test sets.

B.3: Results and Evaluation

Quantitative Results

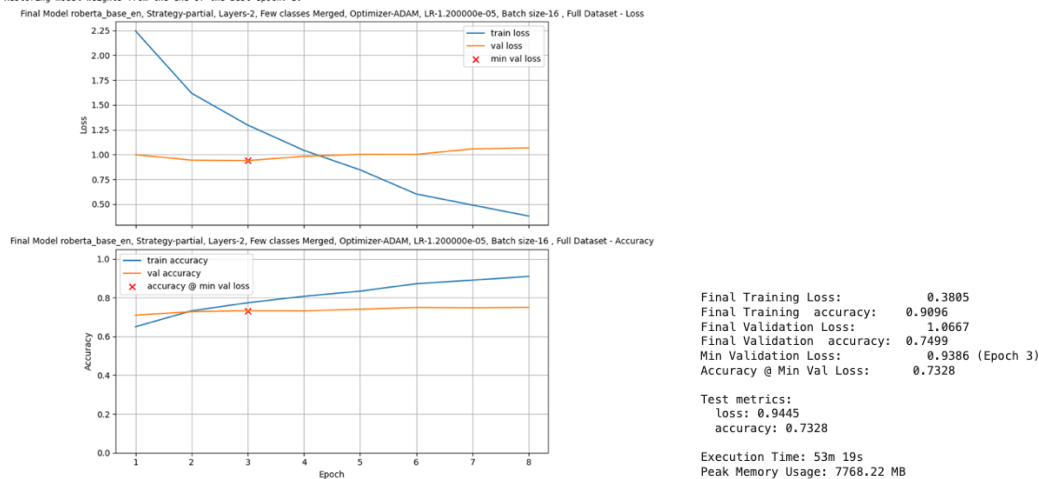
The final RoBERTa-based classification model, trained with a partial fine-tuning strategy of the last 2 transformer backbone, achieved a Test Accuracy of 73.28% and a Test Macro-F1 score of 0.6605. The training process exhibited rapid convergence, identifying optimal weights at epoch 3. At this point, the model achieved its minimum Validation Loss of 0.9386 and a Validation Accuracy of approximately 75%. The final training logs reveal a significant divergence between the Final Training Accuracy (90.96%) and the validation metrics, where the validation loss eventually rose to 1.0667 by the end of the run. This trajectory confirms the successful intervention of the Early Stopping callback, which restored the model to its optimal state at epoch 3 to maximize generalization capability.

Interpretation of Results

The performance metrics reveal distinct behaviors across the 28 consolidated categories. The model performed best on categories with distinct, domain-specific vocabulary, such as STYLE & BEAUTY (F1: 0.865), FOOD & DRINK (F1: 0.842), and TRAVEL (F1: 0.838),. Conversely, the model struggled significantly with broad, abstract categories that lack consistent thematic boundaries; WEIRD NEWS (F1: 0.366) and IMPACT (F1: 0.447) were the hardest to classify correctly. The confusion matrix exposes clear patterns of semantic overlap that persist despite our label engineering efforts. The most significant misclassifications occurred between MEDIA and POLITICS (30.2% error rate), likely because news about media organizations often centers on political coverage. Similarly, COMEDY was frequently misclassified as ENTERTAINMENT (22.7%), and BLACK VOICES was confused with ENTERTAINMENT (17.2%), suggesting that the model struggles to distinguish specific cultural lenses from general entertainment news when the subject matter overlaps. The divergence between the high training accuracy (90.96%) and the lower validation accuracy indicates a tendency overfitting beyond the point of max validation accuracy at minimum validation loss, which is common in large transformer models trained on imbalanced data; exiting at the max validation accuracy at minimum validation loss and the use of class weights and strict regularization (dropout) helped maintain a respectable Macro-F1 score.

Figure 3-5: Illustrates the train v/s validation loss and accuracy how it changed with each epoch of training, the F1 Macro, and the test accuracy obtained with the final model.

Model	Strategy	Unfreeze Layers	DS_Merged	Batch	Optimizer	Learning Rate	Val Acc @ Min Loss	Min Val Loss	Test Loss	Test Acc	Test F1 Macro	Elapsed Sec	Peak Memory(MB)	Dataset Fraction
roberta_base_en	partial	2	Few classes Merged	16	ADAM	0.000012	0.732843	0.938625	0.944483	0.732807	0.66049	3199.5	7768.2 MB	1.0



Summary of Final Model Performance

This RoBERTa model represents our "best" solution because it successfully leverages transfer learning to capture the nuances of short text sequences better than our initial baselines. By optimizing the architecture to unfreeze only the top 2 layers using an Adam optimizer with a learning rate of $1.2e-5$ and a batch size of 16 (as selected by Optuna), we achieved a balance between computational efficiency and accuracy. The results indicate that while the task of categorizing news is inherently difficult due to subjective labels (e.g., distinguishing "Parents" from "Parenting"), the model provides a robust baseline. A surprising strength was its ability to achieve high accuracy on visually distinct categories like "Style," while its primary weakness remains the disambiguation of thematically adjacent topics like "Weddings" and "Divorce" (14.2% confusion).

B.4: Discussion and Reflection

Limitations and Sources of Error

The primary limitation of our project stems from the inherent semantic ambiguity and subjectivity within the HuffPost dataset itself. Despite merging of categories, distinct boundaries remain elusive for semantically overlapping topics; for instance, the model frequently mis-

classifies MEDIA as POLITICS (30.2% error rate) and COMEDY as ENTERTAINMENT (22.7% error rate). This indicates that the single-label classification constraint is fundamentally restrictive for news articles that naturally belong to multiple categories simultaneously. Furthermore, the extreme class imbalance—where "POLITICS" contains over 32,000 records compared to fewer than 1,000 for "EDUCATION"—persisted as a source of bias. While class weights helped, the model still exhibited a tendency toward overfitting, evidenced by the divergence between the final training accuracy (90.96%) and the validation accuracy (~75%). The model had high capacity to memorize the training data, but the short, noisy nature of headlines limited its ability to generalize perfectly to unseen text.

Potential Improvements

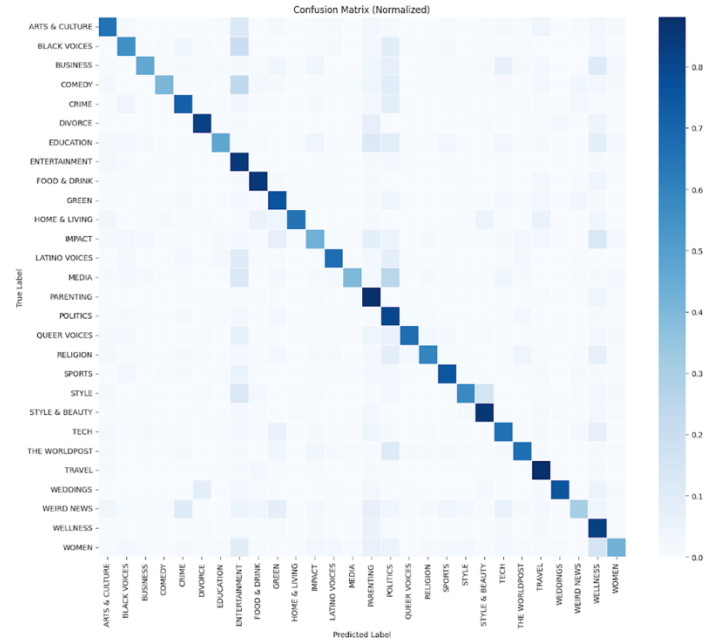
To address these limitations, three improvements could be implemented in future iterations. First, shifting the problem from multiclass to multi-label classification (using a sigmoid activation function instead of softmax) would allow the model to validly predict an article as both "Politics" and "Media," significantly reducing the perceived "error" rate on overlapping topics. Second, implementing Focal Loss instead of standard Cross-Entropy loss would force the model to focus more on hard-to-classify examples rather than the easily classified majority classes, potentially improving performance on minority categories like "Latino Voices". Third, employing data augmentation techniques specifically for minority classes, such as back-translation or synonym replacement, would artificially increase the support for these categories (though it may inject noise), providing a more robust alternative to simple class weighting. We could potentially look at hierarchical classification of semantically overlapping labels.

Reflection on Project Learnings

Our project highlighted that label engineering and data preprocessing were the most significant drivers of performance along with proper hyper parameters identification. We discovered that aggressive text cleaning (removing stop words) in Milestone 2 actually hindered the model, whereas preserving punctuation and sentence structure in the final pipeline allowed the RoBERTa transformer to leverage its pre-trained linguistic knowledge effectively. Regarding optimization, we learned that partial fine-tuning (unfreezing only the top 2-3 layers) offered the ideal balance between training stability and adaptation; training the full model did not result in good classification scores, while freezing it entirely yielded poor accuracy. If we were to repeat the project, we could implement hierarchical classification to first predict broad topics (e.g., "Lifestyle") and then narrow down to specifics (e.g., "Wellness" vs. "Healthy Living") to better manage the dataset's inherent semantic hierarchy.

Conclusion

Our project successfully addressed multiclass classification challenges by integrating pretrained RoBERTa-model utilizing transfer learning with strategic label engineering. By partially fine-tuning the top 2 transformer layers, the final model achieved a robust Test Accuracy of 73.28% and Macro-F1 of 0.6605, with Early Stopping at Epoch 3 proving essential to mitigate overfitting as training accuracy rapidly reached 90.96%. Ultimately, this pipeline offers a viable solution for automated news categorization, demonstrating that resolving ambiguous labels, such as "Parents" vs. "Parenting", is as critical to model performance as the architecture itself.



B.5: AI Use Disclosure

Our team utilized Generative AI tools, specifically ChatGPT, to help with technical implementation challenges, such as identifying best practices for traversing nested KerasNLP layers to implement partial fine-tuning correctly. It was also useful for generating boilerplate code for Optuna visualizations and formatting markdown. However, the tool proved unreliable for tasks; for example, it failed to troubleshoot meeting recording permissions because it could not account for student account restrictions, it fabricated URLs, and it occasionally "hallucinated" optimization strategies or deleted critical information from responses. AI was unable to perform core strategic tasks like Label Engineering, where deciding which categories to merge (e.g., "Taste" into "Food & Drink") required manual interpretation of TF-IDF scores and confusion matrices. We advise future students to treat AI as a syntax helper rather than a project manager. It is excellent for debugging complex loops, but a human must strictly verify the output, especially for complex architectures, as it lacks the context to define data strategy and may fabricate logic or omit essential steps in error.

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Appendix:

GitHub Project Folder:

https://github.com/Siobhan-C/Module_7_Final_Team_Project

AI Use Disclosure:

Our team used AI tools, as described within our paper, such as formatting in markdown, APA, troubleshooting, and other functions. Supporting documentation can be found in our GitHub project folder.